

INDUSTRY PROJECT

Name of the Company

Tata Consultancy Services (TCS)

A REPORT ON

E-Commerce Sales Insights Dashboard

Name of the Institute:

P.E.S College of Engineering, Mandya – 571405

Project Duration:

6 April, 2026 – 25 May, 2026

Total Effort:

90 Hours

Project Environment:

Python, VS Code

Tools Used:

Python, Pandas, Flask, HTML, CSS, JavaScript, Chart.js, FPDF

Date of Submission:

28 April, 2026

Submitted By

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ABSTRACT

The increasing growth of e-commerce platforms has led to the generation of large volumes of sales data, making it essential for businesses to analyze this data effectively for better decision-making. The importance of this project lies in its ability to convert raw transactional data into meaningful insights that help organizations understand sales trends, customer behavior, and profit patterns.

This project implements an **E-Commerce Sales Insights Dashboard** using a structured methodology that includes data collection, data preprocessing, analysis, and visualization. The backend is developed using **Python and Flask**, where data is processed using Pandas and exposed through RESTful APIs. The frontend is built using **HTML, CSS, JavaScript, and Chart.js**, enabling interactive dashboards with real-time filtering capabilities.

The major results obtained from the system include identification of key performance indicators such as total sales, total profit, average order value, and order count. The dashboard also highlights important insights such as the top-performing region (West), the most profitable product category (Technology), and the negative impact of high discounts on profit margins. The system successfully demonstrates how business intelligence tools can improve data-driven decision-making and provide valuable insights for business growth.

Acknowledgement

I would like to express my sincere gratitude to my organization, TCS ION, for providing me with the opportunity to work on the project titled “*E-Commerce Sales Insights Dashboard* ”. This project has been a valuable learning experience and has helped me enhance my practical knowledge in the field of data analysis and data visualisation.

I would also like to thank my mentors and guides for their continuous support, valuable suggestions, and guidance throughout the project. Their insights played a crucial role in shaping the outcome of this work.

I extend my sincere thanks to my institute, P E S College of Engineering, for providing the necessary resources and academic support required to complete this project successfully.

Finally, I would like to thank everyone who directly or indirectly contributed to the completion of this project.

Bheemaray

1.Introduction

In today's digital economy, e-commerce businesses generate massive amounts of data through transactions, customer interactions, and product sales. Efficient analysis of this data is crucial for improving business strategies and maximizing profitability. This has led to the adoption of **Business Intelligence (BI)** systems, which transform raw data into meaningful insights through analysis and visualization.

The title of this project, "**E-Commerce Sales Insights Dashboard**," includes several important terms.

- **E-Commerce Sales** refers to online buying and selling transactions.
- **Insights** refer to meaningful patterns or conclusions derived from data.
- **Dashboard** is a visual interface that displays key information such as charts and KPIs (Key Performance Indicators).
- **Business Intelligence Application** refers to a system that collects, processes, and analyzes data to support decision-making.

This project uses modern technologies such as **Python, Flask, Pandas, HTML, CSS, JavaScript, and Chart.js** to develop a web-based dashboard that presents sales data in an interactive and user-friendly manner. The system allows users to filter data based on parameters such as date range, region, and category, and visualize trends using charts and graphs.

Related work in this domain includes traditional data analysis tools like Microsoft Excel and advanced BI tools such as Power BI and Tableau. While these tools provide strong analytical capabilities, they are often either limited in interactivity or require licensing and advanced configuration. In contrast, this project provides a lightweight, customizable, and cost-effective solution tailored for small to medium-scale applications.

Overall, the project demonstrates how data analytics and visualization techniques can be integrated into a web-based system to provide actionable insights, thereby improving business performance and supporting informed decision-making.

1.1 Problem Statement:

In e-commerce, a large amount of sales and customer data is generated every day. But it is difficult to understand this raw data and get useful insights from it. Traditional methods like Excel sheets and manual reports take a lot of time, may contain errors, and do not provide real-time analysis.

Because of this, businesses face problems such as:

- Difficulty in finding sales trends
- Difficulty in comparing regions and product categories
- Not clearly understanding how discounts affect profit
- Delay in making proper business decisions

So, there is a need for a system that can quickly analyze data and show results in a simple and interactive way.

Therefore, this project develops a **web-based dashboard** that converts raw sales data into clear insights using charts, filters, and reports, helping businesses make better decisions.

1.2 Objectives

- To collect and preprocess sales data
- To analyze data and extract key performance indicators (KPIs)
- To develop a backend system using Python and Flask
- To design an interactive frontend dashboard
- To implement filtering functionality
- To generate reports in multiple formats
- To identify business insights and patterns
- To validate results with original dataset

2. Software Tools

The development of the E-Commerce Sales Insights Dashboard involves the use of various software tools and technologies for data processing, backend development, frontend design, and visualization.

1. **Python**

Python is used as the main programming language for the project. It handles data processing and backend logic. In simple terms, it is used to write the main code of the project.

2. **Pandas**

Pandas is a Python library used for data analysis. It is used to load the dataset, clean the data, and perform calculations. In simple terms, it is used to analyze and manipulate data.

3. **Flask**

Flask is a lightweight web framework in Python. It is used to create backend APIs that connect the data with the frontend. In simple terms, it acts as a bridge between data and the user interface.

4. **HTML (HyperText Markup Language)**

HTML is used to design the structure of the webpage. In simple terms, it creates the layout of the dashboard.

5. **CSS (Cascading Style Sheets)**

CSS is used for styling the webpage. It helps in designing the dashboard appearance, including features like dark and light mode. In simple terms, it makes the dashboard look attractive.

6. **JavaScript**

JavaScript is used to add dynamic behavior to the webpage. It fetches data from backend APIs and updates the dashboard in real time. In simple terms, it makes the dashboard interactive.

7. **Chart.js**

Chart.js is a JavaScript library used for data visualization. It helps in creating charts such as line charts, bar charts, and pie charts. In simple terms, it converts data into graphs.

8. **VS Code (Visual Studio Code)**

VS Code is the development environment used for writing and managing the project code. In simple terms, it is the tool used to develop the application.

9. **FPDF / ReportLab**

These libraries are used for generating PDF reports. In simple terms, they help in creating downloadable reports from the dashboard.

10. **Web Browser (Chrome/Edge)**

A web browser is used to run and test the application. In simple terms, it is used to view the dashboard.

3. Working Procedure

The working procedure of the E-Commerce Sales Insights Dashboard describes how the system processes raw data and converts it into meaningful insights through different stages.

Step 1: Data Collection

The process begins with collecting the sales dataset, which contains information such as order details, customer data, product categories, sales amount, profit, and discount. This dataset is stored in a CSV file and acts as the input to the system.

Step 2: Data Loading

The dataset is loaded into the system using the Pandas library in Python. The data is stored in a structured format called a DataFrame, which allows easy manipulation and analysis.

Step 3: Data Preprocessing

The loaded data is cleaned and prepared for analysis. This includes removing duplicate records, handling missing values, correcting data formats (especially dates), and standardizing category names. This step ensures that the data is accurate and consistent.

Step 4: Data Processing and Analysis

The cleaned data is analyzed to extract useful information. Calculations are performed to determine key performance indicators such as total sales, total profit, number of orders, and average order value. The data is also grouped by month, region, and category to identify trends and patterns.

Step 5: Backend Processing (Flask APIs)

A Flask-based backend is used to process user requests. When the user selects filters such as date, region, or category, the frontend sends a request to the backend. The backend processes the request, applies the required filters to the dataset, and returns the results in JSON format.

Step 6: Frontend Interaction

The frontend dashboard receives data from the backend and displays it using charts and graphs. Technologies like HTML, CSS, and JavaScript are used to build the interface, while Chart.js is used for visualization. The dashboard updates dynamically based on user inputs.

Step 7: Data Visualization

The processed data is presented in different graphical formats such as line charts for sales trends, bar charts for category comparison, donut charts for regional distribution, and scatter plots for profit versus discount analysis. This helps users easily understand the data.

Step 8: Filtering and User Control

Users can apply filters such as date range, region, and product category. Based on these

inputs, the dashboard updates the results in real time, allowing users to analyze specific data segments.

Step 9: Report Generation

The system provides an option to download reports. Users can export the filtered data in CSV format or generate a summarized PDF report. This feature is useful for sharing and documentation.

Step 10: Output and Insights

Finally, the system displays key insights such as top-performing regions, best-selling products, and the impact of discounts on profit. These insights help businesses make informed decisions.

4. MATERIALS AND METHODS

MATERIALS AND METHODS

The development of the E-Commerce Sales Insights Dashboard follows a structured and systematic approach, divided into multiple phases. Each phase focuses on a specific part of the project, ensuring clarity, modularity, and efficient implementation.

1. Phase 1: Data Collection and Understanding

In this phase, the dataset used for the project is collected and analyzed to understand its structure and contents. The project uses the Superstore Sales Dataset, which contains transactional data from 2018 to 2021. The dataset includes important attributes such as Order ID, Order Date, Ship Date, customer details (ID, Name, Segment), product details (Category, Sub-category, Product Name), sales, quantity, discount, profit, and region, state, and city. This phase helps in understanding the nature of the data and identifying key variables required for analysis.

2. Phase 2: Data Cleaning and Preprocessing

Raw data often contains inconsistencies, missing values, and errors. Therefore, preprocessing is performed to ensure data quality and reliability. The methods applied include conversion of date fields into standard datetime format, removal of duplicate records using unique identifiers, handling missing or null values in categorical fields, standardization of category and product names, and validation of numerical fields such as sales, profit, and discount. This phase ensures that the dataset is accurate, consistent, and ready for analysis.

3. Phase 3: Data Transformation and Analysis Logic

In this phase, the cleaned data is transformed into meaningful formats suitable for analysis and visualization. The dataset is loaded using the Pandas library in Python and stored in a DataFrame for easy manipulation. The transformation includes grouping data by month to analyze sales trends, aggregating data by region and category, and calculating key performance indicators such as total sales, total profit, average order value, and total number of orders. The analysis logic includes calculating monthly sales trends using date-based grouping, analyzing regional performance using sum of sales per region, studying the profit versus discount relationship to identify business impact, and identifying top products based on highest sales. These transformations convert raw data into structured insights.

4. Phase 4: Backend Development

The backend of the system is developed using Python and Flask, which acts as a server to process requests and provide data to the frontend. The methods used include RESTful API development, data processing using Pandas, and JSON formatting for data exchange. The key API endpoints include `/api/kpis` for overall performance metrics, `/api/sales_by_month` for monthly sales data, `/api/top_products` for top-performing products, and `/api/sales_by_region` for

regional sales distribution. The backend handles data processing and ensures efficient communication between data and the user interface.

5. **Phase 5: Frontend Development**

The frontend is responsible for displaying data in a visual and interactive format. The technologies used include HTML for structure, CSS for styling and layout design, JavaScript for dynamic content and API interaction, and Chart.js for data visualization. The features implemented include interactive charts such as line, bar, donut, and scatter plots, dynamic filtering based on date range, region, and category, dark mode and light mode user interface, and a responsive dashboard layout. The frontend fetches data from backend APIs and renders it into meaningful visual representations.

6. **Phase 6: Report Generation**

The system provides functionality to generate reports in different formats. The methods used include CSV export using Pandas and PDF generation using FPDF or ReportLab. The features include downloading filtered data, generating summary reports, and providing business insights in report format. This phase enhances usability by allowing users to export and share results.

7. **Software and Tools Used**

The tools used in this project include Python as the core programming language, Pandas for data manipulation and analysis, Flask for backend API development, HTML and CSS for frontend structure and design, JavaScript for dynamic behavior and API calls, Chart.js for data visualization, VS Code as the development environment, and FPDF or ReportLab for report generation.

8. **Project Plan (Step-by-Step Process)**

The overall workflow of the project includes defining project objectives and scope, collecting and studying the dataset, performing data cleaning and preprocessing, transforming and analyzing data to extract insights, developing backend APIs using Flask, designing the frontend dashboard using web technologies, integrating backend and frontend components, implementing data visualization using Chart.js, adding filtering and report generation features, and testing and validating the system outputs.

5. Challenges & Opportunities

Challenges

- 1. Handling raw data with missing values, duplicates, and inconsistent formats**
Raw datasets often contain incomplete or repeated entries and different formats. Identifying and correcting these issues is necessary to ensure reliable analysis.
- 2. Cleaning and preprocessing data to ensure accuracy**
Data cleaning involves removing errors, filling missing values, and standardizing formats. This step improves the quality of data and ensures correct results.
- 3. Designing efficient data processing for fast results**
The system must process large datasets quickly without delays. Efficient algorithms and optimized data handling are required for smooth performance.
- 4. Managing backend and frontend integration using Flask APIs**
The backend and frontend must communicate properly through APIs. Designing these APIs correctly ensures data is transferred without errors.
- 5. Ensuring smooth data communication between server and dashboard**
Data should be sent and received without delays or loss. Proper request and response handling is important for real-time updates.
- 6. Selecting appropriate charts for clear data visualization**
Different types of data require different charts for better understanding. Choosing the right visualization helps users interpret data easily.
- 7. Maintaining responsiveness and user-friendly interface**
The dashboard should work smoothly on different devices and screen sizes. A simple and clean interface improves user experience.
- 8. Implementing dynamic filtering (date, region, category)**
Users should be able to filter data based on their needs. Implementing this feature requires proper handling of inputs and real-time updates.
- 9. Generating accurate reports in CSV and PDF formats**
Reports must reflect correct data and proper formatting. Ensuring accuracy during export is important for documentation and sharing.
- 10. Debugging errors during development and testing**
Errors can occur during coding and integration. Identifying and fixing these issues is necessary to ensure the system works correctly.

Opportunities

1. Integration of real-time data instead of static dataset
2. Implementation of machine learning for sales prediction
3. Deployment on cloud platforms for scalability
4. Adding user authentication and role-based access
5. Improving mobile responsiveness of the dashboard
6. Enhancing visualization with advanced charts and features
7. Expanding the system for large-scale business applications
8. Applying the system to other domains like healthcare and finance
9. Improving performance for handling large datasets
10. Adding more advanced analytics and decision-support features

6. Reflections on the Project

1. This project provided valuable hands-on experience in developing a complete end-to-end system, starting from raw data processing to building an interactive dashboard. It helped in understanding how data can be transformed into meaningful insights that support business decision-making.
2. Through this project, knowledge of data analysis using Pandas, backend development using Flask, and frontend design using HTML, CSS, and JavaScript was strengthened. It also improved skills in data visualization using Chart.js and handling real-world datasets with preprocessing techniques.
3. One of the key learnings was the importance of clean and structured data for accurate analysis. The project also highlighted the challenges of integrating multiple technologies and ensuring smooth communication between different components of the system.
4. Additionally, the project enhanced problem-solving skills, debugging ability, and understanding of system design and workflow. Overall, this project was a practical learning experience that bridged theoretical knowledge with real-world application and prepared for future work in data analytics and software development.

7. RESULTS AND COMPARISON

The developed E-Commerce Sales Insights Dashboard successfully generates meaningful analytical results from the input dataset. The system processes transactional sales data and presents key performance indicators (KPIs) along with visual insights.

1. Predicted / Generated Results

The dashboard produces the following major outputs:

- **Total Sales:** ₹2,297,200.86
- **Total Orders:** 5,010
- **Total Profit:** ₹286,397.02
- **Average Order Value:** ₹458.52

In addition to KPIs, the system provides analytical insights such as:

- The **West region** generates the highest revenue among all regions
- The **Technology category** contributes the highest share of total sales
- The **Consumer segment** yields the highest profit
- A consistent increase in sales trend is observed over the years
- Higher discounts (above 20%) negatively impact profit margins

Graphical outputs include:

- Monthly sales trend (line chart)
- Sales distribution by region (donut chart)
- Top 10 products (bar chart)
- Sales by category (bar chart)
- Profit vs Discount relationship (scatter plot)

2. Comparison and Validation with Original Data

The predicted results generated by the dashboard are validated against the original dataset to ensure correctness and reliability.

- The **aggregated values (sales, profit, orders)** match the calculations derived directly from the raw dataset
- The **monthly trends** align with the original transaction records when grouped by date
- The **regional and category-wise distributions** are consistent with the dataset's actual values
- The **profit vs discount relationship** observed in visualizations matches the numerical data patterns

Validation is performed using:

- Cross-verification with raw dataset values
- Recalculation using Pandas aggregation functions
- Manual comparison of sample records

The results confirm that the system accurately processes and represents the data without loss or distortion.

3. Performance Observation

- The system responds efficiently to user queries and filters
- Real-time updates are achieved through API integration
- Visualization improves understanding compared to raw data tables

Conclusion

The E-Commerce Sales Insights Dashboard successfully demonstrates the application of data analytics and business intelligence techniques in transforming raw data into actionable insights. The project achieves its objective of providing an interactive and user-friendly platform for analyzing sales performance.

Key Findings

- The West region is the highest revenue-generating region
- The Technology category contributes the most to total sales
- High discounts significantly reduce profit margins
- Sales show an overall increasing trend over time
- Consumer segment plays a major role in profitability

8. Picture description

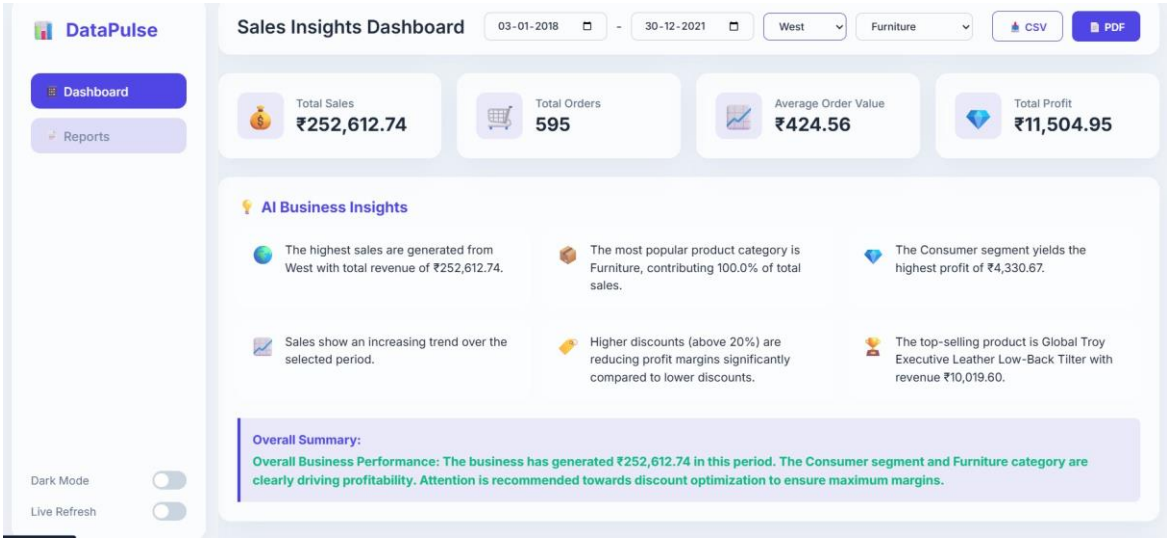


Fig 7.1: Frontend Data Flow Diagram of the E-Commerce Sales Insights Dashboard

The data flow for the E-Commerce Sales Insights Dashboard, illustrated in Fig 7.1, begins with raw data that is preprocessed and analyzed using Python and Pandas. The core communication is managed by a Flask-based backend, which exposes the analytical data through RESTful APIs in JSON format. The frontend, developed using HTML, CSS, JavaScript, and Chart.js, drives the process by sending user requests (e.g., filters for date or region) to the backend. Upon receiving the JSON response, the frontend dynamically updates the dashboard, converting the structured data into interactive charts and graphs for visualization and report generation.



Fig 7.2 : E-Commerce Sales Insights Monthly Sales Trends

Figure 7.2 illustrates the E-Commerce Sales Insights Monthly Sales Trends, typically visualized using a line chart to show performance over time. This trend is generated by grouping the transactional sales data by month. The analysis reveals a key finding of the project: a consistent increase in sales is observed over the years, demonstrating overall business growth. This capability is crucial for identifying seasonal patterns and supporting forecasting decisions.

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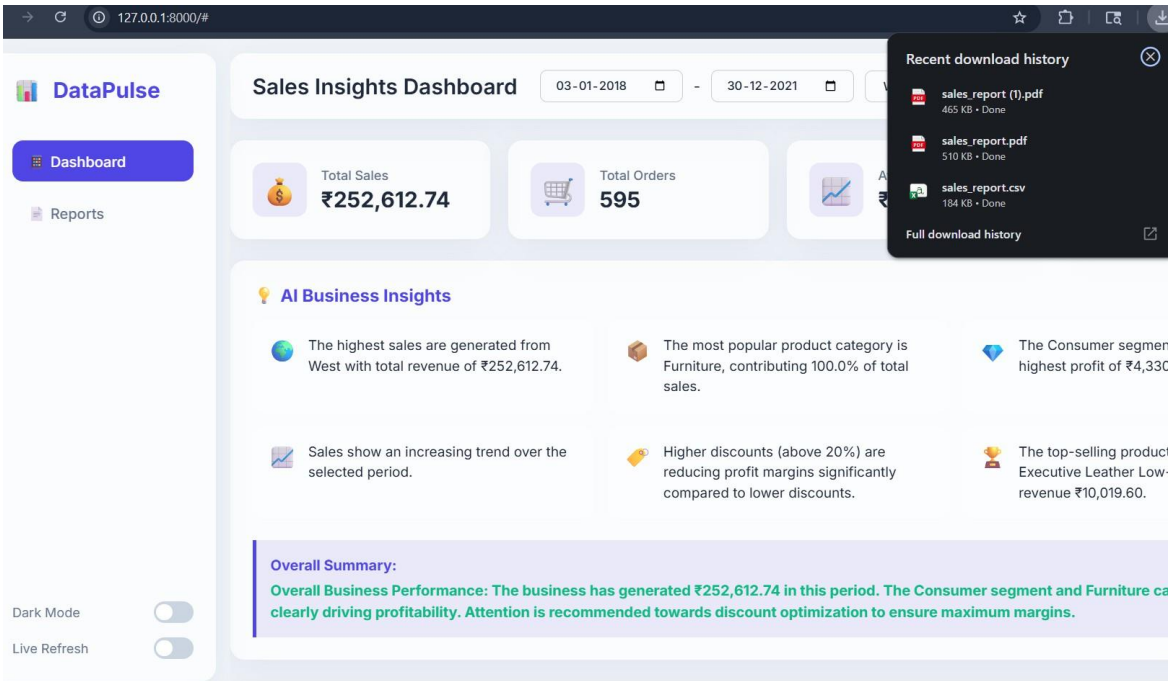


Fig 7.3 Downloading of a CSV and a Sales_report

The system provides report generation functionality, allowing users to export filtered data in CSV format (using Pandas) or generate a summarized PDF report (using FPDF or ReportLab) for sharing and documentation.

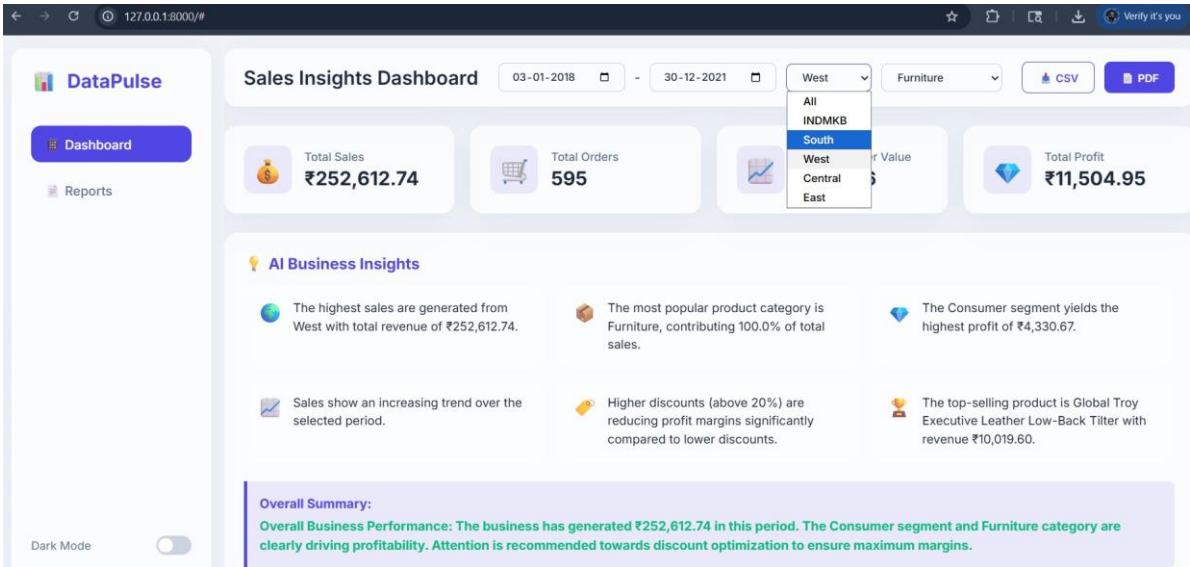


Fig 7.4 : Selecting a different region Sales

The dashboard includes a dynamic filtering feature that allows users to select specific regions, such as the West region, to analyze regional performance distribution. When a user selects a different region, the frontend sends a request to the Flask-based backend, which processes the input and updates the interactive charts and KPIs in real time. This

functionality enables businesses to identify top-performing areas and understand geographical sales trends effectively.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	State	Postal Code	Region	Product ID
1	INDMK8	8-11-2020	11-11-2020	Standard Class	as	asa	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798
2	CA-2020-152156	8-11-2020	11-11-2020	Standard Class	CG-12520	asasa	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454
3	CA-2020-138688	12-6-2020	16-6-2020	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240
4	US-2019-108966	11-10-2019	18-10-2019	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760
5	US-2019-108966	11-10-2019	18-10-2019	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760
6	CA-2018-115812	9-6-2018	14-6-2018	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487
7	CA-2018-115812	9-6-2018	14-6-2018	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833
8	CA-2018-115812	9-6-2018	14-6-2018	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275
9	CA-2018-115812	9-6-2018	14-6-2018	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-100013910
10	CA-2018-115812	9-6-2018	14-6-2018	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892
11	CA-2018-115812	9-6-2018	14-6-2018	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10000539
12	CA-2018-115812	9-6-2018	14-6-2018	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002033
13	CA-2021-114412	15-4-2021	20-4-2021	Standard Class	AA-10480	ssdsd	Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-10002365
14	CA-2020-181389	5-12-2020	10-12-2020	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-100013656
15	US-2019-189883	22-11-2019	26-11-2019	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-10002311
16	US-2019-189883	22-11-2019	26-11-2019	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-BI-10000756
17	CA-2018-105893	11-11-2018	18-11-2018	Standard Class	PK-19075	Pete Kriz	Consumer	United States	Madison	Wisconsin	53711	Central	OFF-ST-10004186
18	CA-2018-167164	13-5-2018	15-5-2018	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	84084	West	OFF-ST-10000107
19	CA-2018-143336	27-8-2018	1-9-2018	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-AR-10003056
20	CA-2018-143336	27-8-2018	1-9-2018	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	TEC-PH-10001949
21	CA-2018-143336	27-8-2018	1-9-2018	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-BI-10002215
22	CA-2020-137330	9-12-2020	13-12-2020	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AR-10000246
23	CA-2020-137330	9-12-2020	13-12-2020	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AP-10001492
24	US-2021-156909	16-7-2021	18-7-2021	Second Class	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-CH-10003174
25	CA-2019-106320	25-9-2019	30-9-2019	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	FUR-TA-10000577
26	CA-2020-121755	16-1-2020	20-1-2020	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	OFF-BI-10001634
27	CA-2020-121755	16-1-2020	20-1-2020	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TEC-AC-10003027
28	US-2019-150630	17-9-2019	21-9-2019	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-BO-10004834
29	US-2019-150630	17-9-2019	21-9-2019	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-BI-10000474
30	US-2019-150630	17-9-2019	21-9-2019	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-FU-10004848
31	US-2019-150630	17-9-2019	21-9-2019	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-EN-10001509
32	US-2019-150630	17-9-2019	21-9-2019	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-AR-10004042
33	US-2019-150630	17-9-2019	21-9-2019	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-AR-10004042

Fig 7.4 : Data set used for a Both Backend and a frontend

The project utilizes the Superstore Sales Dataset, which encompasses transactional data recorded from 2018 to 2021. This comprehensive dataset includes various attributes such as Order ID, Order Date, Ship Date, and customer details including ID, Name, and Segment. Additionally, it contains product-specific information like Category, Sub-category, and Product Name, alongside numerical fields for sales, quantity, discount, and profit, and geographical data including region, state, and city.